

Anjaneya Bhardwaj

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Introduction

As an aspiring AI/ML Engineer currently pursuing MS in Artificial Intelligence from University at Buffalo, I focus on applying AI, deep learning and reinforcement learning to solve complex, real-world problems. My experience includes designing diverse systems, ranging from multi-modal classification models and generative diffusion architectures for time-series forecasting to exploring novel techniques for maximising reward functions using RL. I am committed to leveraging my skills to develop impactful, data-driven solutions within a collaborative environment.

Education

University at Buffalo, SUNY

2025 - May 2026

MS, Artificial Intelligence

- **Achievements:** Developed skills in Generative Diffusion Models, Neural Network design, Reinforcement Learning and PyTorch through team projects. Led collaborative efforts to build and optimize neural models for predictive tasks, managing data workflows and team coordination to deliver results on time.
- **Coursework:** Basics of AI, Intro to ML, Numerical methods for computing and data science, Data Intensive Computing, Reinforcement Learning, Computer Vision, Pattern Recognition

Jaypee Institute of Information Technology (JIIT), Noida, Sec-62

2020 - 2024

B.Tech, Information Technology

- **Achievements:** Developed skills in problem solving, leadership, time management, product planning, solution delivery, and cross-team collaboration over four years of engineering.
- **Coursework:** Data Mining, Web Algorithms, Information Retrieval, Machine Learning, Big Data, Financial Accounting

Experience

ECHO India | Data Science Intern

Jun 2023 - Jul 2023

- Collaborated with senior data scientists and analysts to collect, clean, and preprocess large datasets from diverse sources using Python and pandas, enabling downstream predictive modeling
- Created visualizations to illustrate key trends in user behavior, helping stakeholders quickly identify areas for improvement
- Performed NLP tasks and built a sentiment analyzer with the VADER lexicon in Python, automatically classifying user feedback and highlighting positive and negative trends for the product team

Certifications

- **Supervised Machine Learning: Regression and Classification:** Issued Nov 2024. Learned about regression for predictive numerical analysis and classification for accurate data categorization for robust decision-making in diverse applications. Online Course by Stanford, DeepLearning.AI.
- **Affective Computing:** Issued May 2024. Gained skills in emotion recognition, computational modeling of emotions, and analysis of multimodal data (voice, facial expressions, physiological signals) using machine learning and signal processing techniques. Developed an understanding of ethical, legal, and social implications in human-machine interaction. Online Course by NPTEL, IIIT-D.

Projects

Multi Agent RL for Network Intrusion Detection | <https://github.com/AnjaneyaBhardwaj/MARL-for-CyberSecurity>

- Leveraged **Multi agent reinforcement learning** and Graph Neural Networks within the **CybORG++** environment to develop an AI driven cybersecurity defense system that can adapt to adversarial attacks while intelligently prioritizing protection of critical assets under resource constraints.
- Modeled a digital enterprise network as a graph and provided the embeddings using **GNN** to a blue (Defender) agent to make it topology aware.
- Implemented adversarial self play using **Proximal Policy Optimisation (PPO)** and compared it to baseline **Double Deep Q-Network** as a baseline metric.
- Demonstrated **47%** improvement over baseline metric.

Deafine- From Sound to Sense | <https://github.com/AnjaneyaBhardwaj/Deafine>

- A simple Python accessibility tool that provides real-time speaker detection and live transcription using **ElevenLabs Speech-To-Text API** for deaf and hard-of-hearing users.
- Implemented realtime audio processing with **WebSocket** and made sure that there were no redundant API calls when there is silence through batched API calls.
- Secured the MLH Best Use of ElevenLabs Award at MLH-UB Hackathon

Multi Modal Genre Prediction | <https://github.com/AnjaneyaBhardwaj/movies-genre-classification>

- The Multi-Modal Genre Classification project aims to improve movie genre prediction by combining textual data (plot summaries) and visual data (movie posters). Using a dataset of over **100,000 movies** sourced from **TMDb API**, we balanced six main genres through cleaning, augmentation, and undersampling.
- Three models were developed: a **Text-only LSTM classifier**, an **Image-only MLP classifier**, and a **Fusion model** that integrates both modalities using a multi-layer neural network.
- Results show that Fusion model with multi-modal learning significantly enhances genre classification performance, with a **72% accuracy**.

Healthcare Data Management and Warehousing System | <https://github.com/AnjaneyaBhardwaj/Data-Mgmt-Warehousing>

- Designed a normalized **PostgreSQL** schema (3NF) to ensure data integrity and minimize redundancy, alongside a dimensional **Star Schema** data warehouse optimized for complex analytical queries.
- Built an automated pipeline using **Python** to ingest raw datasets and utilized **dbt (data build tool)** to engineer Fact and Dimension tables, enabling reporting on key metrics like hospital revenue and patient length-of-stay.
- Utilised **FastAPI** to expose healthcare data via **REST endpoints**, facilitating seamless integration for external applications and data consumers and containerized the full stack (Database, API, and Pipeline) using **Docker** and **Docker Compose**, ensuring consistent development and deployment environments.
- Enforced data protection standards by implementing **Role-Based Access Control** within the database, defining granular permissions for analysts and application users. Furthermore, implemented **performance tuning** by adding **indexes** to the database that improved the execution time from **616 ms to 53 ms (about 90%)**.
- Finally carried out the data analysis to address important business questions on patient demographics, doctor workload and billing.

Vital Sign Forecasting using Diffusion Models | <https://github.com/AnjaneyaBhardwaj/Vital-sign-forecasting-using-Diffusion>

- The main motivation was to not just get the most likely outcome, we wanted to visualise the full distribution of trajectories. In this project we treated vital sign forecasting not as a regression task, but as a **Generative AI "In-Painting" task**, which gives us **50 different trajectories**, instead of just one.
- Our core contribution is a Conditional Denoising Diffusion Probabilistic Model (**DDPM**) adapted for medical time series. We implemented a 1D **ResNet-style** architecture. It uses 1D convolutions to process temporal dependencies and a sinusoidal embedding to encode the diffusion time- steps.

- Finally, compared our implementation to a baseline LSTM, that showed the how LSTM only gives one trajectory of the vitals whereas our implementation gives a cloud of a potential trajectories.

Leadership and Activities

Light De Literacy NGO

Oct 2020 - 2024

- Taught underprivileged children and contributed to their health and well-being.
- Served as Admissions Head, facilitating the admission process for children.
- Organized fundraising activities, led cloth donation drives, and coordinated cultural events.

Pre-Startup Competition | *Prognosis.AI*

- Led a three-member team on outlining the problem statement for a pre-startup idea initiative by the college. Our aim was to design a platform for doctors and patients alike to ensure continuity of care and early detection of lifestyle diseases. Our efforts led the idea to reach the final stage.

Skills

- Machine Learning, Plotly, Seaborn, Neural Networks, Data Analysis, Pytorch, Reinforcement Learning, Artificial Intelligence, Generative AI, Data Visualization, FastAPI, APIs, Diffusion Models, Docker, Python, TensorFlow, PostgreSQL, SQL, LLMs, RAG, Scikit learn, Statistical Modelling, Data Manipulation, Data Mining, Git, LangChain, Pandas, ETL, NLP, Data Pipelines, Gen AI, Gym Library, Gymnasium